

Mansi Mantri

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of internship experience designing and deploying distributed systems, RESTful APIs, and backend services using Python, Java, Node.js, and AWS. Strong foundation in data structures, algorithms, and system design, with hands-on exposure to ML frameworks and LLMs.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6+), TypeScript, SQL, Kotlin

CS Fundamentals: Data Structures, Algorithms, System Design, OOD, Debugging, Unit & Integration Testing (Jest, PyTest, JUnit)

Distributed Systems & Cloud: AWS (S3, Lambda, Step Functions, IAM), Docker, Kubernetes (basics), GitHub Actions, CI/CD, Microservices, Kafka, RabbitMQ

Backend: Node.js, Express.js, FastAPI, Spring Boot, REST & GraphQL API Design, JWT/OAuth

Databases & Caching: PostgreSQL, MongoDB, MySQL, Supabase, Firebase, Prisma ORM

ML & LLMs: LLMs (DeepSeek, CodeT5+, Llama 3), Hugging Face, RAG Pipelines, AWS Bedrock

Tools: Git, Linux, Agile/Scrum, Postman, System Design

EXPERIENCE

VODA.ai | Data Analyst Intern

Jun 2026 - Aug 2026

- Queried and maintained SQL relational databases, performing large-scale cleanup, deduplication, and standardization across 10K+ records to support accurate data collection and reporting.
- Built automated reporting dashboards using Power BI and HubSpot, improving data visibility and reducing manual reporting effort by 50%.
- Developed data-driven enrichment scripts and API-based hooks to automate scoring workflows, translating business requirements into functional reporting tools.

Dottir Labs | Software Engineer Intern

Feb 2026 - Apr 2026

- Studied the existing system and redesigned the legacy Tkinter desktop app into a modern web application using React.js, FastAPI, and Julia (for data processing), with Plotly.js dashboards for spectral data visualization.
- Implemented core features including spectral data processing, peak detection with manual override, calibration curve generation, and offline/online connectivity with an offline upload queue and retry logic, plus Google Drive and local storage integration for CSV/JSON data.
- Conducted end-to-end system testing before pilot deployment, identifying and resolving 15-20 error-handling issues - including battery safety checks, hardware/laser testing, and crash recovery - directly enabling the system's successful advancement to pilot phase.
- Partnered with Design and QA engineers to diagnose production issues, review code, and iterate rapidly on fixes.

Park Pulze | Backend Engineer Intern

Sep 2025 - Dec 2025

- Developed and optimized scalable backend systems handling high-volume data (2K+ users), improving system throughput by 35%.
- Designed secure RESTful APIs with authentication and access control, improving data synchronization performance by 40%.
- Debugged and enhanced system reliability through testing and code reviews, reducing issue resolution time by 25%.

Crossing | Software Engineer Intern

Sep 2025 - Dec 2025

- Built and supported distributed AWS pipelines (S3, Lambda, Step Functions) integrating ML preprocessing, delivering production data processing at 200+ records/week scale.
- Developed backend systems with asynchronous processing and LLM-based validation checks (AWS Bedrock), reducing manual review effort by 60%.
- Built a RAG pipeline for automated document extraction and filing assistance for attorneys, reducing a 2-3 hour manual filing process to 5 minutes.

Atlas Copco | Full Stack Developer

Jan 2024 - Jun 2024

- Architected and shipped a real-time microcontroller reservation platform (React/Next.js, Node.js/Express, PostgreSQL) using WebSockets for conflict-free, login-authenticated scheduling.
- Engineered a containerized deployment pipeline with Docker for one-command deployment; documented APIs via Swagger and delivered a polished UI with Tailwind CSS.
- Drove system reliability through query and API optimization, resolving 30+ production issues and cutting latency by 20%, partnering with a cross-functional team of 10-15 engineers.

Applied Technology Solutions, Inc. | Software Engineer (R&D)

Jan 2023 - Dec 2023

- Resolved UI/UX issues on client-facing portals and optimized legacy web platforms for a 20% increase in system performance and user engagement.
- Built and tested backend services using Java and Spring Boot, using Postman for API testing, and managed SQL database migrations, reducing data processing time by 30%.
- Developed Python automation scripts and Linux-based testing/validation workflows, reducing deployment time by 40% and increasing test coverage by 30%.

PROJECTS

BigDocBot - AI Documentation Tool

- Built a Python-based code summarization ETL pipeline using LangChain and transformer models (CodeBERT, CodeT5) to extract function-level metrics across large codebases.
- Constructed an AST-based parser to generate line-by-line LLM explanations and token-based heatmaps, making complex code easier to understand and document.

Arena Vibes - Ticket Booking System

- Built scalable RESTful APIs with Node.js and PostgreSQL supporting real-time booking workflows, optimizing 30+ SQL queries and reducing response latency by 20%.
- Developed end-to-end full-stack architecture with TypeScript enforcement, debugging complex logic to reduce booking errors by 90% and improve reliability by 40%.

EDUCATION

Worcester Polytechnic Institute | MS, Computer Science | Aug 2024 - May 2026

MGM's Jawaharlal Nehru Engineering College | Bachelor's, Computer Science and Engineering | Aug 2020 - Jun 2024